# ME3160 Power & Refrigeration Cycles

*End Semester Term Project*

## Project – 1 Surrogate-Assisted Optimization of a Methane-Fired Combined Cycle

### 1.1 Background

A gas turbine burning natural gas (modelled as pure methane) rejects exhaust stream at 500–650 °C. Rather than discarding this energy, a combined-cycle plant routes it through a heat recovery steam generator (HRSG) to raise steam for a bottoming Rankine cycle. Real plants are limited not by thermodynamics alone but by materials: turbine blades, HRSG tubes, and the stack all have temperature ceilings or floors that cap what the designer can do. Your task is to build a physics-based model of such a plant, respect those limits, and use data-driven optimization to find the operating point that maximizes overall efficiency.

### 1.2 System Description

*Methane Fired Open Brayton Cycle – Topping Cylce*

- Compressor: ambient air (1.013 bar, 298 K, 21% $O_2$ / 79% $N_2$ by mole) compressed to pressure ratio $r_p$, isentropic efficiency $\eta_c = 0.86$
- Combustor: methane (LHV = 50.0 MJ/kg) burned with excess air at constant pressure. Assume complete combustion (products: $CO_2$, $H_2O$, $O_2$, $N_2$). Combustor efficiency $(\eta_f)$ is 0.98. The fuel–air ratio $\left(\frac{f}{A}\right)$ sets the turbine inlet temperature (TIT)
- Turbine: expands products to 1.05 bar, isentropic efficiency $\eta_t = 0.89$
- No blade cooling air is required as the TIT is below the material limit

Ideal gas law with temperature-dependent specific heats for each species (use $Cp(T)$ polynomials in Borgnakke & Sonntag, Table A.6). Compute mixture enthalpy and entropy from mass-fraction-weighted species properties. Isentropic processes must be solved via the relative-pressure/entropy-function approach $\left(s_2 - s_1 = ((s^o_{T2} - s^o_{T1}) - R\ln\left(\frac{P_2}{P_1}\right),\ s^o_T = \int_{T_o}^{T}\frac{Cp}{T}dT\right)$, not with a constant $k$. Students who use constant $Cp$ will lose marks on the model-fidelity criterion.

*Single-pressure Low Pressure Rankine Cycle – Bottoming Cycle*

- HRSG: exhaust gas from gas turbine heats feedwater through economizer → evaporator → superheater. Gas-side pressure drop 3 kPa. Steam drum pressure is $P_s$ and superheat outlet temperature is $T_s$
- Steam turbine: isentropic efficiency $\eta_{st} = 0.85$, exhausting to condenser at pressure $P_{cond}$
- Condenser: saturated liquid out
- Feed pump: isentropic efficiency 0.80

Steam side properties are to be estimated using CoolProp (CoolProp.CoolProp.PropsSI with fluid 'Water'). No steam tables should be used

## 1.3 Design Variables and Ranges

| Variable | Symbol | Range |
|---|---|---|
| Compression ratio | $r_p$ | 8 – 30 |
| Turbine inlet temperature | TIT | 1200 – 1500 K |
| Steam drum pressure | $P_s$ | 10-60 bar |
| Superheat temperature | $T_s$ | Saturation + 20 K -> HRSG limit |
| Condenser pressure | $P_{cond}$ | 0.05 – 0.15 bar |

## 1.4 Practical Constraints

Some of the hard constraints that cannot be violated are:

i. Turbine inlet temperature (TIT) should be less than 1550 K, corresponding to uncooled nickel superalloy with TBC margin
ii. Gas entering the HRSG should be less than 650 °C, corresponding to carbon-steel tube limit. This couples $r_p$ and TIT
iii. Gas temperature leaving HRSG should be higher than 100 °C to stay about water dew point temperature
iv. $\Delta T_{pinch} \geq 10$K at the evaporator and approached $\Delta T \geq 5$K at the superheater
v. Steam turbine vapor quality is greater than 0.88 to prevent blade erosion
vi. Equivalence ratio in the gas turbine combustor should be less than 0.5 for reduced NOx emission

## 1.5 Objective

Maximize the overall thermal efficiency for a plant sized to handle air mass flowrate $(\dot{m})$ of 100 kg/s

$$\eta_{cc} = \frac{\dot{W}_{GT,net} + \dot{W}_{ST,net}}{\dot{m}_f Q_{LHV}}$$

Report specific work, bottoming-to-topping work ratio, and HRSG effectiveness at the optimum point

## 1.6 Required Methodology

### *Part A – Direct Model*

Write a Python code for the combined cycle that takes inputs as $r_p$, TIT, $P_s$, $T_s$, and $P_{cond}$ and gives $\eta_{cc}, \dot{W}_{net}$, feasibility with constraint margins. Validate it against the required baseline case as given in "Required Validation Test Case" below, and against a published textbook combined-cycle example. Plot $\eta_{cc}$ vs $r_p$ at fixed TIT for the gas turbine alone and confirm the familiar maximum.

### *Part B – Design of Experiments & Surrogates*

- Sample the 5-D design space with Latin Hypercube Sampling (≥ 300 points, scipy.stats.qmc). Record $\eta_{cc}$ and each constraint margin; keep infeasible points, labelled.
- Fit a surrogate models for $\eta_{cc}$ and for the binding constraint margins: a full quadratic Response Surface (least squares, statsmodels or numpy.linalg.lstsq) with ANOVA to identify which terms matter, or an Artificial Neural Network (scikit-learn MLPRegressor or PyTorch, 2 hidden layers, tanh/ReLU) with an 80/20 train–test split and hyperparameter tuning.
- Analyse $R^2$, RMSE, and parity plots on the held-out set. Discuss where the surrogate fails and why

*Part C – Optimization*

- Optimize the surrogate with a constrained method (using a genetic algorithm via pymoo or DEAP).
- Re-evaluate the surrogate optimum with the direct model. Report the gap. If the gap exceeds 0.2 percentage points of efficiency, perform at least one adaptive refinement (add samples near the predicted optimum, refit, re-optimize).
- Sensitivity study: how does the optimum move if the gas entering the HRSG is relaxed to 700 °C, or tightened to 600 °C? Which constraints are active at the optimum, and what does that tell you about where the materials money should be spent?
- Bi-objective view: maximizing $\eta_{cc}$ alone can drive down net specific work (smaller machinery does less work per unit mass flow, which matters for capital cost). Re-run the GA as a bi-objective search: maximize $\eta_{cc}$ and maximize $\dot{W}_{net}$ simultaneously, subject to constraints given in section 1.4 and extract the Pareto front of non-dominated designs. This front is required in the report (see below)

## 1.7 Required Validation Test Case

Before any optimization work is accepted, each group must demonstrate that their code reproduces a fully worked, hand-traceable baseline. The numbers below are computed directly from the tables in Borgnakke & Sonntag (Tables A.7.1/A.7.2 for air, B.1.2/B.1.3 for steam) so they can be checked with no software at all. This is deliberately split into three independent checks, so a discrepancy can be traced to the specific routine at fault rather than lost in a single combined-cycle number.

*Check-1: Gas Turbine (air-standard) check*

*Using pure air as the working fluid and given that* $T_1 = 300$ *K,* $P_1 = 100$ *kPa,* $r_p = 15$*,* $T_3 = 1400$ *K,* $\eta_c = 0.86$*,* $\eta_t = 0.89$*, you should get*

| State | T (K) | h (kJ/kg) | $P_r$ | Notes |
|---|---|---|---|---|
| 1 | 300.0 | 300.47 | 1.1146 | Table A7.1/A7.2, exact |
| 2s | ≈641.2 | ≈650.81 | 16.72 | $P_{r2} = P_{r1} r_p$ |
| 2 (actual) | ≈694.7 | 707.87 | — | $h_2 = h_1 + (h_{2s} - h_1)/\eta_c$ |
| 3 | 1400.0 | 1515.27 | 361.62 | Table A7.1/A7.2, exact |
| 4s | ≈707.3 | ≈721.47 | 24.11 | $P_{r4} = P_{r3}/r_p$ |
| 4 (actual) | ≈787.7 | 808.77 | — | $h_4 = h_3 - \eta_t(h_3 - h_{4s})$ |

Your code should be able to match the above Brayton cycle state variables obtained using Tables A7.1 and A7.2. Based on these state variables, you should get:

- $w_c = 407.4$kJ/kg, $w_t = 706.5$kJ/kg and $w_{net} = 299.1$kJ/kg
- $q_h = 807.4$kJ/kg, $\eta_{th} = 37.05\%$

You should be able to match the temperatures at $T_2, T_4$ and $\eta_{th}$ to establish that the code is able to solve the isentropic relation with variable $C_p$ for pure air

*Check-2: Rankine cycle v/s Steam Tables*

*Given that* $P_s = 30$ bar, $T_s = 450\ °C, P_{cond} = 7.5\ kPa, \eta_{st} = 0.85, \eta_{pump} = 0.80$

| State | Description | h (kJ/kg) | s (kJ/kg·K) |
|---|---|---|---|
| 5 | Turbine inlet (3000 kPa, 450 °C) | 3344.00 | 7.0833 |
| 6s | Isentropic to 7.5 kPa | 2208.4 | 7.0833 |
| 6 | Actual turbine exit | 2378.8 | — |
| 7 | Condenser exit, sat. liquid @ 7.5 kPa | 168.77 | 0.5763 |
| 8 | Pump exit (actual) | 172.54 | — |

Your code using CoolProp calls should recover the above state points of the Rankine cycle and should get the following results:

- $x_{6s} = 0.848$, actual exit quality $x_6 = 0.919$ satisfies constraint C5 (≥ 0.88)
- $w_{st} = 965.2$ kJ/kg, $w_{pump} = 3.77$ kJ/kg, $w_{net} = 961.4$ kJ/kg
- $q_H = 3171.5$ kJ/kg → $\eta_{Rankine} = 30.3$ %

*Check-3: Energy Balance*

The 1st law energy balance should be satisfied for all your runs:

$$\dot{m}_{fuel} Q_{LHV} \eta_{comb} \approx (\dot{W}_{GT,net} + \dot{W}_{ST,net}) + \dot{Q}_{stack} + \dot{Q}_{condenser} + (small\ losses)$$

*Reporting Requirements*

Each report must include a Validation Table with columns Quantity | Hand value | Code output | % difference | Pass/Fail, covering every quantity in Checks 1–3, plus test_validation.py in the repository that asserts these tolerances automatically. Suggested tolerances: 0.5% for the gas-side check (bounded by the hand-calculation's own interpolation error) and 0.1% for the steam-side check (CoolProp vs. tables should agree tightly). A group that cannot meet these tolerances has an actual bug.

## 1.8 Deliverables

- Python repository with README, requirements, and a single reproducible driver script
- A ppt giving the following:

| # | Section | Required content |
|---|---|---|
| 1 | Schematic | A labelled block diagram of the full combined cycle — compressor, combustor, turbine, HRSG (economizer / evaporator / superheater), steam turbine, condenser, feed pump — with state points numbered consistently with every equation and table used later in the report. |
| 2 | Algorithm | A flowchart or structured pseudocode of the solver: how f/TIT is resolved from the combustor energy balance, how the isentropic relations are evaluated with variable cp, how the HRSG pinch |

| # | Section | Required content |
|---|---|---|
| | | point is located, and how the DOE, surrogate fitting, and optimization loop fit together. |
| 3 | Analysis | Model validation against the Required Validation Test Case, T–s and T–Q diagrams at baseline and optimum, surrogate comparison ($R^2$, RMSE, parity plots), single-objective optimization results, constraint activity, and the HRSG relaxed/tightened sensitivity study. |
| 4 | Pareto plot | The $\eta_{cc}$ vs. $W_{net}$ Pareto front from the bi-objective GA run (Part C), with the single-objective optimum marked on it, and a short discussion of where the group would actually choose to operate and why. |
| 5 | Conclusion | The recommended operating point, its efficiency and net specific work, the binding constraints at that point, and the single biggest lever (design variable or material limit) for improving the plant further. |

### 1.9 Rubric

| Criterion | Weight |
|---|---|
| Thermodynamic model fidelity (variable cp, correct mixture properties, correct pinch treatment) | 25 |
| Validation and error analysis (required test case, all 3 checks, within stated tolerances) | 20 |
| DOE and surrogate quality (both RSM and ANN, honest test-set metrics) | 10 |
| Optimization rigor (constraint handling, multi-start/GA, direct-model verification, refinement) | 10 |
| Physical interpretation and sensitivity study | 15 |
| Report and presentation | 20 |